

ARCMiS: Agentic Repository-Level Code Migration for Small Language Models

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Abstract

We present ARCMiS (Agentic Repository-level Code Migration for SLMs), a harness that migrates whole source codebases from one programming language to another with small language models (SLMs, 27B parameters) served locally on one consumer GPU. The harness is a hierarchical multi-agent system: a tier-1 orchestrator runs a ten-phase, evidence-gated state machine and delegates over a task graph; a static three-tier tree puts config-owned team leads (tier 2) over the specialist roles (tier 3), and each lead runs its own inner loop from a scoped batch brief. A deterministic guard hook inspects every tool call from every tier, and an optional typed-decision judge arbitrates the ambiguous remainder with deny-on-uncertain semantics. Role-agnostic hooks (trace, continuation, per-role output budgets) live in a separate middleware crate on the model-call path. All state lives on a character-capped shared blackboard; snapcompact absorbs context overflow by rendering discarded history to PNG bitmap frames instead of calling an LLM summarizer. Scoring is toolchain-only: the target build and test commands must pass in the produced workspace, and agent self-reports never gate. We report the tuned harness’s final benchmark record from a four-day, 14-iteration optimization campaign: 46+ scored runs over eight problem sets produced two toolchain-verified Rust workspaces, a champion (Go→Rust checkdigit) that compiles and passes 48 tests under the **Hardening** phase, and a nine-run verified lineage on the C→Rust nandc set. Five of the eight sets stalled before producing a compiling workspace, both fresh rotation sets failed under the exact tuned defaults, and we record the binding constraints: output-ceiling burn rate, a 16k-token Discovery wall, and set selection. The results establish what a 27B local model can and cannot do at repository scale, and no more.

1 Introduction

Repository-level code migration—translating an entire codebase from one language to another while it keeps its behavior—is one of the hardest open tasks for coding agents. It needs three things at once: reading and understanding code that does not fit in one context window, planning a dependency-respecting translation order, and producing code that passes real toolchains. Large frontier models can attempt this with ad-hoc prompting. Small language models cannot. Small models are also the only affordable option for sustained local use: they run on one consumer GPU, cost nothing per token, and keep proprietary source on the machine.

ARCMiS targets exactly this regime. It migrates source codebases with a 27B model served locally through ollama on a single RTX 3090. The design constraint that shapes the whole system is *honesty of scoring*: an agent’s claim of success counts for nothing. A run succeeds only when the

target toolchain compiles the produced workspace and its test command exits zero. LLM-graded proxy scores, when recorded at all, never gate.

This paper makes four contributions:

- A hierarchical multi-agent methodology for repository-level migration with small models: a tier-1 orchestrator, a static config-owned tree of tier-2 team leads over ten tier-3 specialist roles, a ten-phase evidence-gated state machine, and a shared blackboard (Section 3).
- A middleware separation: the role-agnostic model-call hooks (trace, continuation patch, per-role output budgets) live in their own crate under the criterion that a middleware hook never names a role or the orchestration domain; the guard hook stays with the delegation contract (Sections 3 and 4).
- A benchmark record for the tuned harness, closed 2026-09-29 after a four-day, 14-iteration optimization campaign: 46+ scored runs over eight problem sets, two toolchain-verified workspaces, and honest per-set verdicts including the sets that never compiled (Section 6), plus a hierarchy validation run that reached **Migration** with the toolchain green.
- A failure taxonomy for SLM agents at this scale, distilled from the campaign: think-truncation at output ceilings, provider shape rejections, stagnation loops, and ceiling burns, with the limits they impose on generality claims (Section 7).

Figure 1 shows the system. One orchestrator runs the phase machine. One guard hook sits on every tool call from every tier. Three config-owned team leads sit over the ten specialists, who do the work on a shared blackboard; a tier-1 delegation naming a lead hands the batch to the lead’s inner loop.

2 Related Work

We group the related work into four categories: direct prior art on code migration, multi-agent scaffolds that inspire this design, work that justifies the methodology, and the pipeline agents we compare against.

Repository-level code migration. TransCoder [6] translated functions between C++, Java, and Python with unsupervised machine translation and evaluated with unit tests. AlphaTrans [3] is a neuro-symbolic system that translates a whole repository and verifies behavior with schemas and tests. AlphaTrans defines the repository-level, behavior-verified translation problem that ARCMiS attacks with small models instead of a neuro-symbolic pipeline. CodaMosa [8] escaped coverage plateaus by mixing LLM test generation with search-based testing. Our validator–tester loop applies that lesson with a per-module cap. Our problem sets reuse the ReCodeAgent dataset’s project pairs.

Multi-agent scaffolds. ReAct [21] interleaves reasoning traces with tool actions and is the base loop of most single-agent migration scaffolds. Reflexion [17] adds verbal self-reflection across retries. Self-Refine [12] iterates a generate–feedback–revise loop. ARCMiS borrows the iterate-on-failure shape but moves the feedback to deterministic judges and toolchain evidence, because 27B models self-assess poorly. ReAct and Reflexion define the reasoning-plus-action and retry-with-feedback loops that every migration scaffold in this space instantiates; ARCMiS keeps the loops but replaces model-graded feedback with toolchain evidence. MetaGPT [2] assigns a software company’s

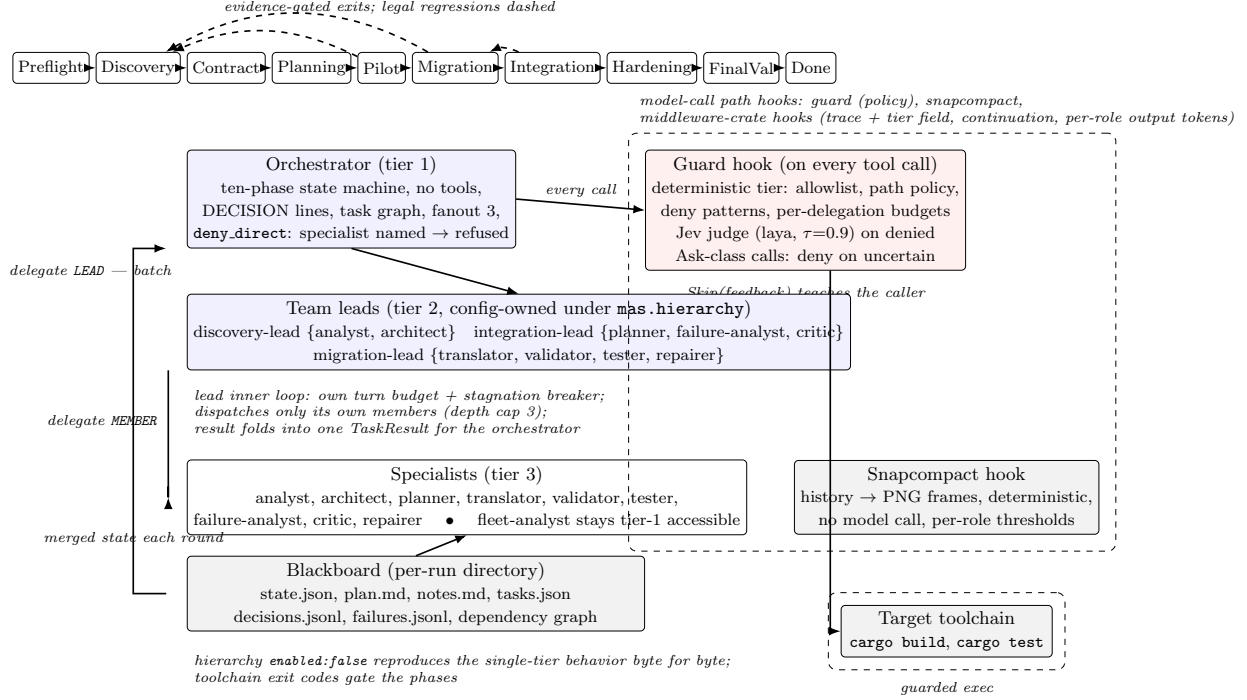


Figure 1: ARCMiS architecture. The tier-1 orchestrator rounds over the ten-phase state machine and emits DECISION lines over a task graph. With the hierarchy enabled, a delegation naming a specialist is refused (**deny_direct**) and the handoff form is a lead name; the lead receives a scoped batch brief and runs its own inner loop over its members (tier 3). The guard hook inspects every tool call from every tier; denied Ask-class calls may arbitrate through the Jev judge (deny on uncertain). The middleware crate holds the role-agnostic hooks (trace, continuation, per-role output budgets); snapcompact attaches to every role’s call path. The target toolchain returns exit codes to the evaluator.

roles to LLM agents over standardized artifacts. AutoGen [18] builds applications from conversing agents with human-in-the-loop checkpoints. CAMEL [10] studies role-playing communicative agents. MetaGPT, AutoGen, and CAMEL assume strong models and free-form conversation; ARCMiS keeps the role decomposition but replaces conversation with a parsed decision protocol, a static registry, and a state machine, because conversation overhead is what small turn budgets cannot afford. SWE-agent [20] shows that a well-designed agent-computer interface matters as much as the model; that interface lesson is the basis for our hashline file tools and guard hook. Agentless [19] shows that a simple localizer-repairer-validator pipeline beats complex agents on some tasks; the campaign’s small-project verdicts agree in spirit, since its two verified sets are the smallest C and Go sets.

Methodology justification. SWE-bench [5] anchored evaluation in executable tests instead of model judgments, and we follow the same principle with toolchain-only scoring: an agent pass counts only when the repository’s own tests pass. Meta-Harness [13] optimizes an executable harness through an agentic outer loop where the proposer reads raw execution traces from the filesystem; the four-day campaign behind our benchmark record follows that contract, with immutable per-run directories and recorded lineage. Toolformer [16] teaches models to call APIs

self-supervised. Gorilla [14] fine-tunes models for API invocation. ARCMiS uses neither: the 17-tool surface is fixed, and the guard, not the model, enforces the calling discipline, which a 27B model cannot learn reliably within its turn budget. DeepSeek-R1 [1] and Qwen3 [15] make deliberate reasoning (`think` blocks) standard on open models at our scale; those reasoning blocks consume output-token budgets and drive the truncation failure mode in Section 7. Speculative decoding [9] makes local inference fast enough for long agent runs, which is why the run budgets of this paper fit on one GPU.

Hierarchical orchestration. Zhu et al. [22] survey multi-agent orchestration frameworks and name hierarchical dynamic-adaptive orchestration as an emerging pattern: static agents whose selection and composition at runtime is dynamic, under multi-tier control. ARCMiS instantiates that pattern in the small-model regime as a static three-tier tree: agents are static (fixed registry, fixed preambles, fixed tool allowlists), the lead-over-member assignments are config-owned, and the task graph, the router chain, and the lead inner loops make selection dynamic. The survey’s deterministic-fallback and policy-enforcement guidance maps one-to-one onto our guard hook and evidence-gated phases.

Pipeline agents for migration. ReCodeAgent [4] prescribes an analyzer–planner–translator–validator pipeline for repository-level translation and validation, and Ledger [7] prescribes a manager–worker scaffold over a shared task ledger. Both exist as arXiv preprints (arXiv:2604.07341, arXiv:2608.26480); we found no published venue version, and our comparisons follow their prescribed algorithms with that status unchanged. The ReCodeAgent paper’s own comparison set is Skel, Oxidizer, AlphaTrans, and SWE-agent, all re-run by its authors with one frontier LLM (Claude 4.5 Sonnet) over 118 projects; ARCMiS runs the same four problem-set families (crust, oxidizer, alphas, skel) with a 27B local model, but the protocols differ on model, scale, and scoring, so Section 6.4 treats those numbers as context, not a comparison.

3 The ARCMiS Methodology

This section describes the method as it ran in the campaign: the role taxonomy and its tiers, the evidence-gated phase machine, the decision protocol with tiered delegation, the blackboard, the guard, the budgets, and the scoring rule.

3.1 Overview

A run is one migration attempt against one problem set: a source codebase in language L_s and a target language L_t , with a test command defined in the config. The harness (a Rust binary built on rig 0.42) spawns a MAS whose members share one blackboard directory. Control is a static three-tier tree: the orchestrator at tier 1 owns the plan and never writes product code; team leads at tier 2 own one batch or phase slice each; specialists at tier 3 own the work and never re-plan. What is dynamic is selection and composition: the orchestrator may emit several delegations in one round, the task-graph executor orders and gates them, and a failing task can spawn a failure-analyst $\rightarrow$ repairer sub-chain without any phase regression. Table 1 shows the role taxonomy and the tier each role sits at.

3.2 Phase state machine

The run walks ten phases: Preflight, Discovery, Contract, Planning, Pilot, Migration, Integration, Hardening, FinalValidation, Done. Forward transitions are earned. The orchestrator’s `done` verb is accepted only when the phase’s exit condition holds *and* at least one delegation completed in the phase (the `phase_delegations` counter). Regressions are legal and recorded. Pilot and Migration may fall back to Discovery on a missing dependency, Integration to Migration, and so on. Anything else is a programming error, not a model decision. Each phase has a written exit condition in the orchestrator prompt. For example, Contract requires `analysis/brief.md` to exist with a validator pass covering it. FinalValidation requires the configured toolchain command to exit zero on the target workspace.

The phase machine is also the campaign’s diagnostic instrument. Every stall in Section 6 is reported as the phase where the run stopped (Discovery or Contract for most failures), so a reader can see at which capability boundary the harness, not the model, gave up.

3.3 Decision protocol and tiered delegation

The orchestrator answers each round with plain text whose lines carry `DECISION: verbs: delegate ROLE | TASK [| after:t2,t3], replan, escalate, done`. Parsing text instead of expecting tool calls removes a whole failure class of small-model agent runs: malformed tool-call JSON that the ollama daemon rejects with HTTP 500. The orchestrator gets *no tools at all*. The round prompt carries the plan, notes, task list, ledger tail, and state, so the model cannot spend turns re-reading files.

Each `delegate` decision becomes a task node with optional `after:` dependencies (Figure 2). Direct tasks run under the task-graph executor: the ready set (all dependencies done) runs with bounded fan-out (3 in the tuned config). Because ollama serves one model at a time, fan-out interleaves turn streams rather than running parallel processes. With the hierarchy enabled, the `deny_direct` gate refuses a delegation that names a team specialist, and the handoff form is `delegate migration-lead | ...`: the tier-1 round prompt lists the lead names in `ROLE`. Routing is a three-stage chain: the orchestrator’s role hint, then a deterministic keyword pass over the task text, then (as a last resort) the router model. After each round the orchestrator re-scores the task list and sees merged state, so a failing task can spawn a failure-analyst → repairer sub-chain without any phase regression.

The lead inner loop. A tier handoff is context management. The lead receives a scoped batch brief (task text, the run contract, and the plan and notes tails under the existing `plan_cap/notes_cap` caps), not the orchestrator’s history. Its inner loop parses `DECISION: delegate MEMBER | task` lines, dispatches one member per round through the same execution path as tier 1, and ends on `done`, its own turn budget, its own stagnation breaker, or a model failure. The lead dispatches only its own members (depth cap 3: lead names are not roles, so a lead cannot dispatch another lead), and its result folds into one `TaskResult` for the orchestrator. A cross-team or lead-named dispatch draws a logged refusal and counts toward the stagnation breaker. The lead’s turn budget absorbs retry loops, which keeps batch churn out of the tier-1 history.

3.4 Blackboard

All mutable state lives in a per-run directory written by the blackboard crate: `state.json` (phase, counters), `plan.md` and `notes.md` (character-capped at 4000 and 8000), `tasks.json` (task list mirror), `decisions.jsonl` and `failures.jsonl` (ledgers), and a dependency graph. Character

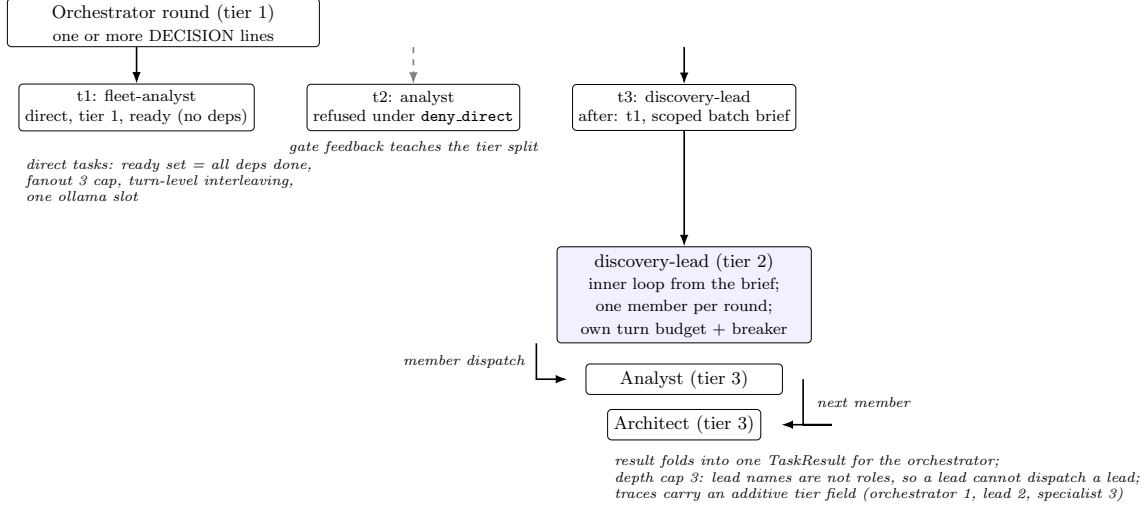


Figure 2: Tiered delegation. The orchestrator (tier 1) emits DECISION lines; the task graph executes direct tier-1 tasks whose dependencies are done, under the fan-out cap. A delegation naming a lead hands the batch to the lead’s inner loop: the lead runs from a scoped batch brief, dispatches only its own members (tier 3), and folds the result into one TaskResult. Delegation events and traces carry an additive tier field (orchestrator 1, lead 2, specialist 3).

caps keep the shared state inside one read for a 16k-token context window; the tuned config raises the per-call window to 65536 with read gating (Section 5), and the caps still bound what the orchestrator prompt itself carries. The writer truncates with an elision marker rather than silently growing the context of every later reader.

3.5 Guard hook and the Jev judge

Every tool call from every agent at every tier passes one `on_tool_call` hook. The deterministic tier of the guard (a policy tier, not a hierarchy tier) inspects the role allowlist (from static tool metadata), path policy (`source/` is read-only, writes must stay inside the workspace), deny patterns for bash (`rm -rf`, `git push`, package installs, network commands, `sudo`), and per-delegation tool-call budgets. Rejections return `Skip(feedback)`: the model sees the reason and adjusts the call, and the breaker counts repeats. The guard lives in the orchestrator crate, beside the delegation contract it enforces; the arbitration seam below it is the only model-policed slot.

Calls that the deterministic tier marks ambiguous (Ask-class) fall to an optional model-arbitration tier built on the `laya` typed-decision crate, following the Jev-as-a-judge pattern [11]. The judge is a small decision-only model that renders one typed `choice` verdict per consultation in a single forward pass, with confidence q defined as the maximum label probability. A confidence cascade with $\tau = 0.9$ accepts confident verdicts; below τ , on any load or inference error, on foreign labels, or with the judge disabled, the guard keeps its deterministic denial. The judge can only widen one denial; it never bypasses a tier that passed the call, and budget denials for non-ask tools are never arbitrable. Deny-on-uncertain is the invariant: the escalation path is the existing deny, because with one model in the loop there is no stronger judge to pay. The deterministic rules stay above the judge: allowlist, path policy, and budgets are untouched.

3.6 Budgets, breaker, fleet ladder

Budgets are per role class: orchestrator turns per round (20), worker turns per delegation (40), judge turns (40, kept small in spirit by prompting judges to promise few calls), lead turns per handoff (**turns** per team in the hierarchy config; the smoke run’s migration-lead worked under a budget of 8), and a run-level round ceiling. A deterministic stagnation breaker stops the run after **stagnation_rounds** (3) consecutive rounds with no completed task, after repeated consecutive phase failures, or when a derived token estimate exceeds the run ceiling; each lead carries its own breaker under the same rule. It never consults a model. In the campaign the breaker was the graceful exit: most stalled runs ended as “3 consecutive rounds without a completed task,” which is a diagnosable artifact, not a hang. The fleet analyst owns a model ladder (weakest first, config-driven) and promotes a role one rung on repeated failure. Per-role output-token and think overrides are config knobs introduced during the campaign (Section 6).

3.7 Scoring

A run succeeds only when the evaluator reruns the target toolchain (**cargo build + cargo test** for Rust targets, **pytest** for Python, or a JavaScript syntax check) in the produced workspace and every test passes. Agent self-reports are recorded but never trusted. This rule exists because self-report traps are real: an agent can write a passing per-problem record over an empty workspace, and a status string carries no signal while an exit code does. The scorer re-runs the toolchain itself, and traces and score paths count tool-call activity, so a healthy tool-only turn (empty text, real tool calls) is never misread as a failure.

4 Supporting Infrastructure

Five components carry the model-call path and the workspace boundary, and one crate boundary keeps them honest. Each exists because a measured failure mode demanded it, and each one is a hook or a pure tool layer, outside agent code.

Middleware crate. The role-agnostic rig **AgentHook** implementations live in their own crate, **ARCMiS/lib/middleware**: the trace hook (one trace record per model call, tool call, and tool result, now carrying the additive tier field), the continuation patch (**TrailingUserMessageHook**), and the per-role output-token hook. The split criterion is that a middleware hook never names a role, a fleet, or the orchestration domain: it reads only hook event data plus its own constructor parameters. Which hook runs for which role, with which thresholds, is domain policy and stays in the agents crate. The guard stays beside the delegation contract in the orchestrator crate, because its allowlist and path policy are the delegation contract, not a generic hook parameter.

Hashline file tools. File tools (**read**, **write**, **edit**) address content through hashline tags (**PATH#TAG**). Edits name the tag they replace. A stale tag is an invalidation, not a silent overwrite. This gives the guard a path-shaped argument to inspect and gives the model a cheap merge-conflict signal.

Snapcompact. When the projected input of the next model call exceeds a per-role token threshold, the snapcompact hook keeps the most recent messages verbatim. It serializes the discarded history to dense text, renders that text to PNG frames with a bundled public-domain 8x13 BDF bitmap font, and attaches the frames as image blocks. The pass is deterministic (same history in,

same frames out) and costs no model call. LLM summarization both spends budget and distorts. The hook lives in its own crate (`snapcompact`); the per-role threshold wiring stays with the registry, because threshold selection is domain policy. It attaches to every role, with lower thresholds for the long-context specialists (translator, validator) that need it most, and `snapcompact` events appear in the traces, which makes context growth observable per agent.

Continuation patch. Ollama’s chat template for Qwen models rejects a message list that ends with tool results and carries no trailing user query (HTTP 500, “no user query found in messages”). `rig 0.42` produces exactly that shape on every multi-turn tool continuation. A completion-call hook appends a synthetic user message (“Tool results above. Continue the task from here.”) to the history. Without it, each rejected call burns a provider retry of 15–55 s.

Tool sandbox. Tool executions run against snapshot-gated workspaces, so the harness can roll back a bad write without touching the source tree.

5 Experimental Setup

This section fixes the configuration that produced every number in Section 6: the model and hardware, the eight problem sets, the budgets, and the per-run protocol that the optimization loop followed.

Model and hardware. All reported runs use the 27B reasoning model `smtek/Swift-Qwen3.8-27B:map-k4v` (4-bit `map-k4v` quantization, `think:true`) served locally by ollama on one RTX 3090, fully offloaded. The tuned per-call configuration is `num_ctx 65536` with guard read gating, `max_output_tokens 8192` with translator and architect raised to 16384, temperature 0.2, fanout 3, and a 7200-second wrapper timeout. One earlier configuration point is quoted in Section 6 because the campaign refuted it: an all-roles 16384 output override.

Problem sets. Eight problem sets from the ReCodeAgent dataset directory `tool_projects`, spanning four language pairs and three orders of magnitude in size (Table 3). The campaign’s search runs used five of them; the two `c14` rotation sets ran only under the exact tuned defaults as a validation step, with no per-set tuning.

Budgets. The MAS configs fix `max_rounds 20`, orchestrator turns 20 per round, worker turns 40 per delegation, judge turns 40, and the stagnation breaker at 3 rounds. `plan_cap 4000` and `notes_cap 8000` bound the blackboard documents.

Protocol and campaign loop. One process per run, strictly sequential, fresh workspace per run, model offloaded between runs and verified by polling the daemon. Every run leaves an immutable experiment directory: `manifest.json` (model, budgets, git revision, parents, hypothesis), config, `traces/turns.jsonl` (one line per model call, tool call, and tool result), the produced `workspace/`, captured streams, and a `result/` aggregate with a staged failure taxonomy (`setup`, `analyze`, `plan`, `translate`, `verify`, `evaluate`). Scoring is toolchain-only (Section 3). The optimization loop over these runs kept the problem set and evaluator fixed for all 14 iterations; every change was harness-side. Section 5.1 states the loop’s contract and the per-iteration hypotheses.

5.1 Optimization methodology

The 14 iterations followed a Meta-Harness-style optimization loop [13]: an agentic outer loop whose proposer reads the full experiment history from the filesystem and proposes the next harness change. The loop ran under five contract rules, each enforced by the experiment tree rather than by trust:

- **Immutable per-candidate directories.** One experiment per directory under a dedicated artifacts root. The directory is immutable after the run: the proposer reads, never writes. It carries `manifest.json`, the produced workspace, raw traces (one JSON line per model call, tool call, and tool result), and the scored result.
- **Pre-registered causal hypothesis.** Each manifest records a one-sentence hypothesis before launch, plus `parents` links to prior experiment ids. Lineage lives in the manifests; search scripts never reconstruct it. The proposer must name the mechanism its change targets, so every refuted candidate stays attributable.
- **Toolchain-only rescoring.** The scorer reruns the target build and test commands inside the produced workspace; agent self-reports never gate. `success: true` requires translation to complete and the toolchain test command to exit zero.
- **Pareto frontier.** `frontier.py` reads only the experiment tree and orders candidates by verified pass, per-problem test count, and wall time. A candidate dominates another when it is not worse on every key and strictly better on one. The frontier, not any single score, selects the champion.
- **Converge-or-rotate.** The 3-cycle rule: if the best candidate on a set shows no frontier gain across three consecutive cycles, the set has converged. A converged champion either closes the loop or rotates to fresh, unseen sets under the exact tuned defaults as an out-of-sample check.
- **No-tuning rule.** The problem set and the evaluator stayed fixed for all 14 iterations; every change was harness-side. No run config was edited per set, and no prompt was tuned against a known failure.

The loop found every binding constraint by reading raw traces, not summaries. Table 4 attributes each lineage step to its recorded hypothesis; Figure 4 plots the verified-test lineage it produced.

6 Results

The benchmark record is the final state of the four-day optimization campaign: 14 harness iterations, 46+ scored runs, eight problem sets, closed 2026-09-29. The problem set and evaluator never changed; every iteration changed the harness. Every data path in this section reads from the per-run artifacts. Table 5 gives the per-set verdicts, and Figure 3 shows the best toolchain-verified result per set.

6.1 Champion and lineage

The champion run is the checkdigit candidate closed at **Hardening** on 2026-09-28: 13 rounds, 14 delegations, 12 of 13 tasks done, and a translated Rust crate whose `cargo build` is clean and whose

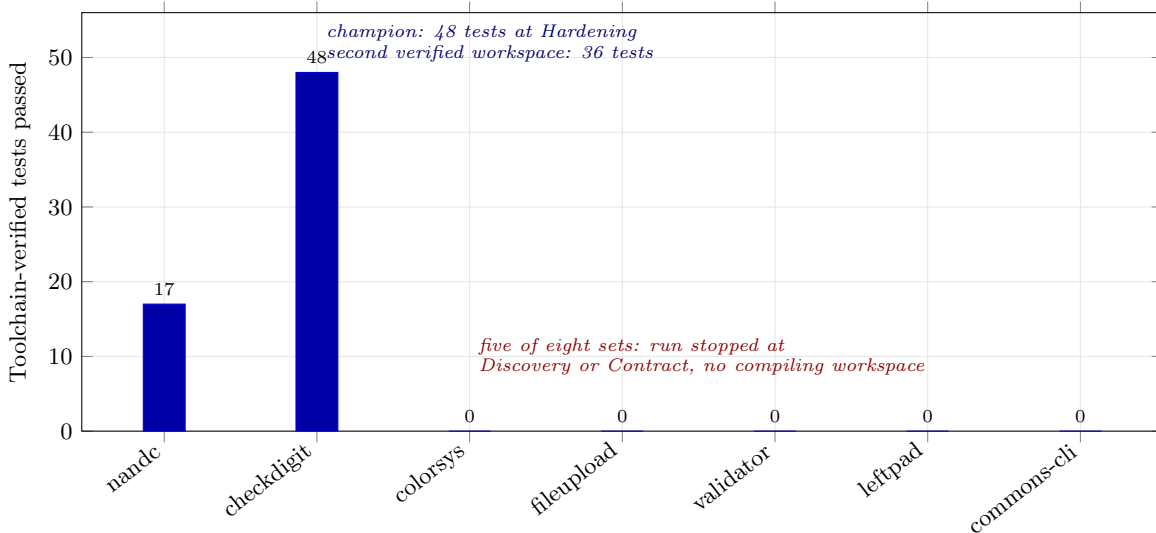


Figure 3: Best toolchain-verified test count per problem set over the whole campaign. Only nandc (17 tests) and checkdigit (48 and 36 tests) produced compiling workspaces; every other bar sits at zero with the run stopped at Discovery or Contract.

test suite passes, toolchain-verified in the produced workspace (12 `cargo test` cases including unit, integration, and doc tests; the campaign scorer’s rescore counts 48 test cases). The run stopped on the wrapper’s runtime timeout mid-**Hardening** after a repairer pass, not on a model failure. Figure 4 shows the lineage: the campaign peak is two toolchain-verified oxidizer workspaces (48 and 36 tests on two checkdigit candidates) and one nandc lineage carrying five generations.

The lineage records what each pre-registered hypothesis bought (Table 4). The seed configuration plus trace hooks produced the first verified workspace (2 tests) once the c1 hypothesis removed the 4096-token think-truncation. The c3 read-scoping and context-raise hypothesis fixed Discovery saturation on multi-file sets and made the stall breaker the graceful exit (17 tests). The c10b per-role output-cap hypothesis carried the architect past Contract, which landed the 48-test champion. The c11b rerun under a doubled timeout looked like progress at 36 tests but was a favorable sample, and the regression test against it failed: the c12 all-roles 16384 override showed that bigger burns cost 10–20 minutes per retry against 1–2 minutes at 8192, and total throughput fell. The c13 revert to the translator-only override showed no depth gain, and the plateau closed the loop under the 3-cycle rule. The c14 rotation hypothesis then failed out of sample: two fresh sets under the exact tuned defaults stalled before Contract.

6.2 Hierarchy validation

After the campaign closed, one smoke run validated the three-tier hierarchy (Section 3) end to end on the GildedRose C→Rust set, with the campaign champion as its parent. The run reached Migration in 13 rounds with 5 of 8 tasks done and a green toolchain: the produced workspace’s `cargo build` and `cargo test` exit zero, and the `text_test` binary reproduces the source’s 31-day walkthrough (day 0 through day 30) line for line. The run is explicitly incomplete (`completed: false`); it stopped on the stagnation breaker after the migration lead exhausted its turn budget of 8 on verbose task text. The delegation record shows the mechanics working as designed: 7 tier-2 hand-offs under the three leads, 19 lead turns, and 14 tier-3 member dispatches, with `discovery-lead`

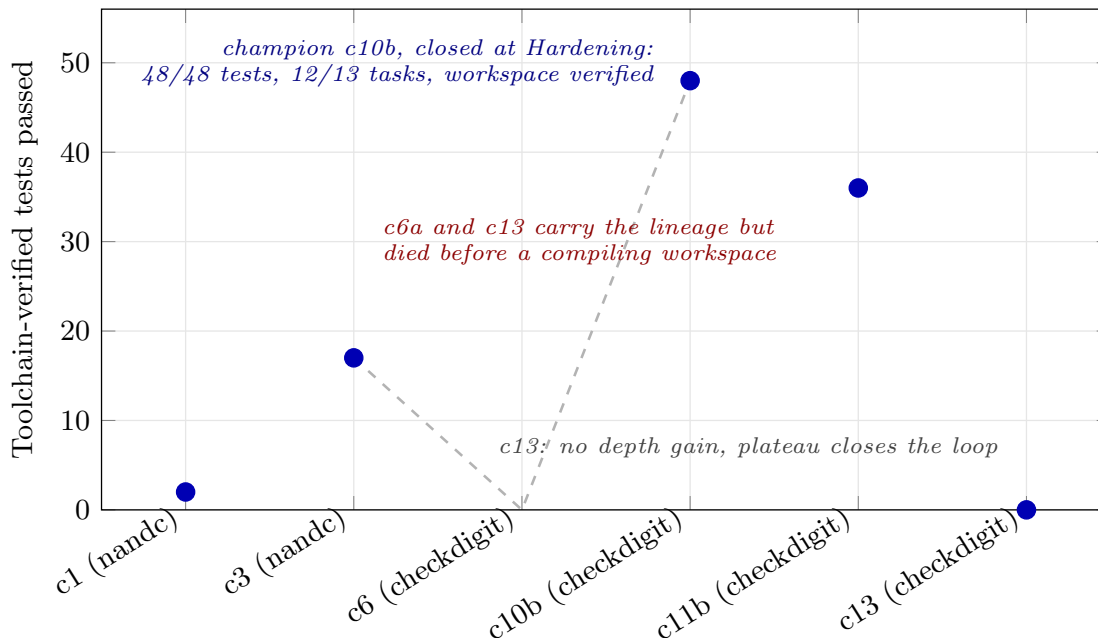


Figure 4: Champion lineage by toolchain-verified tests passed, one point per accepted hypothesis in the optimization loop. The nandc lineage runs from the first verified migration (2 tests at Hardening, after the c1 output-cap fix) to its 17-test generation (c3, after the context raise). The checkdigit lineage peaks at 48 tests (c10b, after the per-role output-cap raise), regresses to 36 (c11b, a favorable sample rather than a config effect), and shows no depth gain at c13, whose plateau closed the loop. Every binding constraint on this path was found by the loop reading raw traces, not by prompt changes.

and `migration-lead` trace records carrying the tier field (tier 2) and their specialists tier 3. This is a validation of the hierarchy mechanics, not a completed migration, and it does not change the campaign verdicts.

6.3 Capability boundaries

The verdicts split the eight sets into three regimes, and the boundary tracks corpus type, not LOC alone. The two verified sets are small, single-purpose C and Go codebases (125 and 1,643 source LOC). The `colorsys` set (355 LOC, Python) cleared `Contract` twice and never a verifying build. The four `alphatrans` sets (Java, 8.6k–36k LOC) never cleared `Discovery` or `Contract`: `fileupload` stalled at `Discovery` across six runs with two of three `Discovery` failures after the gate fix, validator saturated `Discovery` before the gate fix existed, and `commons-cli` stalled at `Discovery` under the final tuned defaults. The `c14` rotation sets are the sharpest evidence: `leftpad` is 82 LOC of C, smaller than the solved `nandc` set, and it still stalled at `Contract` under the exact tuned defaults; `commons-cli` stalled at `Discovery`. Generality of the tuned harness is therefore NOT established. The gains are set-specific to `nandc` and `checkdigit`, and convergence stopped the loop before a third rotation sample could test whether `Discovery`-phase failure on Java sets is a corpus-wide pattern.

The binding constraint is the model’s round-budget burn rate: ceiling burns plus slow artifact turns. No ceiling policy, timeout raise (3600 to 7200 seconds), or context raise (16k to 32k to 64k) moved it. Both wrapper-timeout kills in the campaign landed on runs that were still progressing,

and raising the timeout only moved the wall; the burn rate binds first.

6.4 Baseline comparison: published context

One comparison surface exists in this paper: the published ReCodeAgent record, which serves as context only. ARCMiS never ran Skel, Oxidizer, AlphaTrans, or SWE-agent on shared instances under one protocol, so every cross-paper juxtaposition below is directional, not statistical.

Published ReCodeAgent numbers as context. The ReCodeAgent paper (arXiv:2604.07341, accepted at ASE 2026) benchmarks ReCodeAgent against four techniques, reproduced by its authors with the same LLM: Skel, Oxidizer, AlphaTrans, and SWE-agent, over 118 real-world projects across four PL pairs (C→Rust, Go→Rust, Java→Python, Python→JavaScript), all driven by Claude 4.5 Sonnet with a 5,000-second per-agent budget. Its headline results: 99.4% compilation success; 86.5% test pass rate over validated developer tests against 25.7% for the competing techniques (2,107 validated tests total; per-technique TPR: Oxidizer 67.2%, Skel 93.2%, AlphaTrans 188 of 1,181 executed tests, SWE-agent 78.3% on the Crust subset); and 98.1% on translated developer tests. Table 6 collects these published numbers.

Three protocol differences make these numbers non-comparable to every ARCMiS number in this paper. First, the model: a frontier proprietary model against one 27B quantized model on one GPU. Second, scale: projects averaging 1,975 LOC (up to about 38,000) against our 82 to 36,356, with per-agent budgets of 5,000 seconds and an average cost of \$15.3 per project. Third, validation: ReCodeAgent translates and generates tests in both source and target languages and uses author-validated suites withheld until after the run, whereas a campaign pass requires the pre-registered evaluator commands to exit zero on the produced workspace only. The sets overlap in family (oxidizer/checkdigit, alphas/trans/fileupload and validator, skel/colors/sys, crust), not in instance lists or scoring. We therefore draw no ranking from the juxtaposition. What the comparison does support is directional: the published record shows a multi-agent pipeline with iterative validation outperforming single-shot and neuro-symbolic baselines by wide margins on the same projects, which is consistent with our finding that harness discipline, not model prompting, carries small models past single-agent limits.

7 Discussion

The campaign’s value beyond the verdicts is the failure record. This section names the four mechanisms that dominate it, the limitations that bound every claim, and the threats to how the numbers were read.

Failure taxonomy. Four failure modes dominate the campaign record, and each one names the harness change that addressed it. (1) *Think-truncation*: with `think:true`, reasoning blocks consume the output budget; a turn that dies at exactly the ceiling with empty text and no tool call is a hard completion error, and retries reproduce it deterministically at temperature 0.2. The architect and planner burns at exactly 8192 tokens killed more runs than any other mechanism; per-role output overrides address it. (2) *Provider shape rejections*: ollama’s server-side chat template 500s on message lists without a trailing user query, on tool continuations (patched by the hook, Section 4) and on a first-call request shape with an empty prompt field. (3) *Stagnation*: without a breaker, the orchestrator retries the same delegation shape indefinitely. The 3-round breaker converts an unbounded loop into the campaign’s most common stop reason and its graceful exit.

(4) *Ceiling burns*: an output ceiling kills a role only when the whole artifact cannot fit, but the retry cost depends on where the ceiling sits: 1–2 minutes at 8192, 10–20 at 16384. Total wrapper throughput, not per-turn survival, is the metric, which is why a translator-only override dominates an all-roles override, and why the all-roles policy was refuted as a global default.

Limitations. Three limits bound every claim in this paper. First, rotation: both fresh sets under the exact tuned defaults stalled before Contract, so generality is not established, and any new harness idea must first clear Discovery on a Java alphasets set before it can claim otherwise. Second, convergence stopped the loop at two rotation samples, so the question of whether Discovery-phase failure on Java sets is corpus-wide is open. Third, one model, one quantization, and one GPU dominate the data; the ceiling-policy finding in particular must be re-derived per model, not assumed to transfer. Toolchain-only scoring can still pass a workspace whose tests are weaker than the source suite; the tester’s per-module generated-test cap bounds but does not eliminate that gap.

Threats to validity. Sample sizes are small (single scored runs on the rotation and validator sets), so the per-set verdicts are existence and boundedness claims, not rates, and we make no significance claims. The search sets and the harness changes saw each other for four days; the c14 rotation is the only out-of-sample evidence, and it failed. Daemon and infra failures contaminate raw counts in the earlier records; the experiment contract gates on daemon liveness, and traces count tool-call activity so that a healthy tool-only turn is not misread as a failure.

8 Conclusion

ARCMiS shows that repository-level code migration with a small local model is tractable only when the harness carries most of the discipline: a tiered orchestrator with evidence-gated phases, a deterministic guard on every tool call with deny-on-uncertain arbitration, character-capped shared state, toolchain-only scoring, and a stagnation breaker that turns silent loops into artifacts. The final benchmark record is deliberately honest: two verified Rust workspaces (48 and 36 tests) on the champion set, a five-generation verified lineage on nandc, and five of eight sets that never produced a compiling workspace, including both fresh rotation sets under the exact tuned defaults. The durable findings are harness findings: the binding constraint is the model’s burn rate, not any single timeout or context size; throughput, not per-turn survival, is the metric for output policy; and the next harness idea must clear Discovery on a Java set before it may claim generality. The infrastructure (the immutable experiment contract, the lineage in the manifests, the failure taxonomy) is what makes every next hypothesis cheap to test and every claimed success impossible to fake.

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Role	Tier	Responsibility
Orchestrator	1	Owns plan and task list. Emits DECISION lines. No tools.
discovery-lead	2	Runs Discovery and Contract over {analyst, architect}.
migration-lead	2	Runs Planning, Pilot, and Migration over {translator, validator, tester, repairer}.
integration-lead	2	Runs Integration and Hardening over {planner, failure-analyst, critic}.
Analyst	3	Reads the codebase. Writes the source map and graphs.
Architect	3	Writes the migration contract (analysis/brief.md).
Planner	3	Produces dependency-ordered translation batches.
Translator	3	Writes target-language modules for one batch.
Validator	3	Judges one translated batch. Re-runs the suite. Read-only.
Tester	3	Adds tests. Runs the test command.
Failure-analyst	3	Classifies one failure into a five-category taxonomy.
Critic	3	Runs the end-of-run adversarial review. Read-only.
Repairer	3	Fixes one diagnosed failure.
Fleet-analyst	1	Model-ladder promotion advice; stays tier-1 accessible (no artifacts, no isolation value).

Table 1: The role taxonomy and tiers. Tier 1 is the orchestrator, tier 2 the team leads, tier 3 the specialists. The lead set is config-owned (**mas.hierarchy**); with the hierarchy disabled the run is byte-identical to the single-tier design. The **deny_direct** gate refuses a tier-1 delegation that names a specialist, so the handoff form under hierarchy is the lead name. The preflight validator rejects a member that sits in two teams and a lead without a prompt file.

#	Phase	Exit evidence	Legal regressions
0	Preflight	deterministic gates, no model call	—
1	Discovery	source map + brief cite every module	—
2	Contract	<code>brief.md</code> + validator pass	—
3	Planning	every module in one batch, order compiles	—
4	Pilot	one batch translated, verified, tested	Pilot→Discovery
5	Migration	every batch translated and verified	Migration→Discovery
6	Integration	full test suite green	Integration→Migration
7	Hardening	critic verdict is pass	Hardening→Integration
8	FinalValidation	toolchain rerun exits zero	FinalVal.→Integration
9	Done	—	—

Table 2: The phase state machine. Forward moves are evidence-gated. Regressions are recorded, not hidden.

Problem set	Pair	Source LOC	Campaign role
crust/nandc	C→Rust	125	search
oxidizer/checkdigit	Go→Rust	1,643	search
skel/colorsyst	Python→JavaScript	355	search
alpatrans/commons-fileupload	Java→Python	8,646	search
alpatrans/commons-validator	Java→Python	36,356	search
crust/leftpad	C→Rust	82	rotation (c14b)
alpatrans/commons-cli	Java→Python	11,643 (52 files)	rotation (c14a)
smoke (GildedRose)	C→Rust	7	diagnostic only

Table 3: The eight campaign problem sets (LOC measured on source trees, headers excluded for C; commons-cli counted as Java files). Test commands come from the per-set config, `cargo test` for Rust targets. The five search sets carried the optimization loop; the two rotation sets validated the tuned defaults on fresh inputs; the smoke set is a diagnostic.

Cycle	Hypothesis (from the manifest)	Outcome
c1	think-only turns truncate at exactly the 4096-token output cap (<code>finish_reason=Length</code> , empty text); raise <code>max_output_tokens</code> to 8192	First verified migration: nandc workspace passes 2 tests. Diagnosis came from raw traces, where every dead delegation died at exactly the cap.
c3	31 of 41 stalled turns sat at the 16384 <code>num_ctx</code> wall; double context, add tool-aware traces for triage	Discovery saturation fixed; nandc reaches 17 verified tests.
c10b	checkdigit’s architect died at exactly 8192 output tokens; raise the architect’s per-role output cap to 16384	Architect passes Contract on the first try; 48-test champion.
c11b	identical-config rerun under a doubled 7200s wrapper timeout; a completed 13/13 would be the first	36 tests; killed by the timeout 6 min after its last pass.
c12	all-roles 16384 output override	Refuted: bigger burns cost 10–20 min per retry against 1–2 min at 8192; total wrapper throughput, not per-turn survival, is the metric.
c13	revert to the translator-only 16384 override; reproduce or beat c11b’s depth	No depth gain; plateau closes the loop.
c14	two fresh sets under the exact tuned defaults, no per-set tuning	Both stall before Contract; out-of-sample generality not established.

Table 4: Lineage steps and the pre-registered hypotheses that caused them, quoted from the per-run manifests. The loop, not the prompt, found each binding constraint: output-cap truncation (c1), the context wall (c3), per-role output budgets (c10b), and the ceiling-throughput trade (c12–c13).

Set	Runs	Final verdict
crust/nandc	9	SOLVED: champion lineage, 17 tests verified
oxidizer/checkdigit	6	CONVERGED: two verified workspaces (48, 36 tests), plateau
alphatrans/commons-fileupload	6	PARKED: stuck at Discovery, ceiling burns unrecoverable in budget
alphatrans/commons-validator	1	NOT CLEARED: Discovery saturation (pre-gate fix)
alphatrans/commons-cli	1	NOT CLEARED: Discovery stall under tuned defaults (c14a)
crust/leftpad	1	NOT CLEARED: Contract stall under tuned defaults (c14b)
skel/colorsys	2	NOT CLEARED: Contract ceiling
smoke (GildedRose)	1	Diagnostic only

Table 5: Final per-set verdicts of the campaign. “Runs” counts scored runs against the set. Three of eight sets produced at least one compiling workspace; none reached **Done** end to end.

Technique	Published result	Protocol
ReCodeAgent	99.4% CS, 86.5% TPR	Claude 4.5 Sonnet, 118 projects
Skel	93.2% TPR (its benchmark)	reproduced with the same LLM
Oxidizer	67.2% TPR (its benchmark)	reproduced with the same LLM
AlphaTrans	188/1,181 tests passed	reproduced with the same LLM
SWE-agent	78.3% TPR (Crust subset)	reproduced with the same LLM

Table 6: Published ReCodeAgent-record numbers [4]. These are the paper’s own protocol: its projects, its validated developer suites, its frontier model. ARCMiS never ran Skel, Oxidizer, AlphaTrans, or SWE-agent on shared instances under its protocol, so this table is context, not a comparison, and no cross-paper statistical comparison with any ARCMiS number is valid.